

# Conceptual Topography

A Field-Theoretic Account of Conceptual Landscapes

VED  $\times$  IFGT Application Layer 1

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*This paper operates within the VED $\times$ IFGT framework established in Endoh [2026b,c,d,e,a]. All variables conform to the Master Variable Table v2.1. The present work does not redefine  $C$ ,  $I$ ,  $\Phi$ ,  $J_I$ ,  $J_C$ ,  $\sigma$ , or  $\tau$ ; it applies them to the domain of conceptual structure and thought dynamics. The primary definition  $\Phi = K * I$  is used throughout; the linear approximation  $\Phi \approx -\alpha C + \Phi_0$  is employed only where explicitly stated. Semantic, psychological, or intentional interpretations of these quantities are not assumed unless explicitly stated. The strict inequality  $C < C_{\max}$  is maintained at all times. Conceptual-domain uses of  $I$ ,  $C$ , and  $\Phi$  should be read as projected or coarse-grained realizations of the IFGT variables within a cognitive observation window.*

## Abstract

This paper presents Conceptual Topography (CT), a field-theoretic account of conceptual landscapes, as the first application layer of the VED $\times$ IFGT framework. A conceptual landscape is defined as the long-time geometry of the information potential field  $\Phi(x, \tau)$  under recursive token-field interaction: a stable, sedimented structure that constrains the trajectories of thought without being directly accessible to conscious inspection.

Within the VED $\times$ IFGT framework, conceptual structure emerges from three coupled processes. First, the continuous information field  $I(x, \tau)$ , inherited from the VED $\times$ IFGT framework as a cognitive-window projection of  $I = f(1 - C)$ , generates an effective potential  $\Phi = K * I$  that shapes the passability landscape. Second, discrete language tokens  $\ell_i = \mathcal{Q}[I, \Phi, C; \text{constraints}]$  act as re-addressable discrete units that interact locally with  $\nabla\Phi$ , deepening potential wells through repeated reuse. Third, the recursive loop  $\Delta \rightarrow C \rightarrow I \rightarrow \Phi \rightarrow J_I \rightarrow \Delta$  drives irreversible sedimentation, producing the attractor geometry  $\Phi \xrightarrow{\text{sed.}} \Phi^*$  that constitutes a mature conceptual landscape.

The theory yields precise structural definitions: a concept is a locally stable region of  $\Phi$  satisfying  $\nabla\Phi(x) \approx 0$ ; understanding is a temporary reduction in traversal cost across such a region; thought is the trajectory of  $J_I$  across the landscape. The strict positivity  $I(x, \tau) > 0$  excludes complete conceptual closure as an ordinary dynamical state. In nontrivial generated landscapes, residual asymmetry and boundary structure allow further gradients and differentiations to reappear.

At the social scale, the same dynamics produce a shared potential geometry maintained across agents through linguistic synchronization, corresponding to the emergence of norms, institutions, and collective intelligence.

This work does not redefine any variable of the VED $\times$ IFGT framework; it applies the existing field-theoretic structure to the domain of cognition and language.

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# 1 Introduction

This paper operates within the VED×IFGT framework [Endoh, 2026b,a] and applies it to the domain of conceptual structure and thought dynamics. All variables follow the Master Variable Table v2.1. The framework developed here is called Conceptual Topography (CT); the phrase “conceptual terrain” refers to the same object when the intuitive metaphor is emphasized.

## 1.1 The Problem of Conceptual Structure

What is a concept? The standard answer places concepts in the semantic domain: a concept is a meaning-bearing unit, a mental representation, or a node in a propositional network. This answer, while useful for many purposes, leaves the *dynamical* question unanswered. How does a concept form, stabilize, and constrain subsequent thought? How does understanding arise, and why does it sometimes fail to transfer even when the relevant definitions are available?

The present paper approaches these questions from a different direction. Rather than asking what a concept *means*, we ask what structural conditions allow a region of the information field to persist, be re-entered at low cost, and systematically bias future dynamics. This reframing shifts the problem from semantics to geometry.

## 1.2 Position in the VED×IFGT Series

The VED×IFGT framework consists of two foundational layers. Vortical Enclosure Dynamics (VED) describes the generation of structure from difference, gradient, flow, and closure [Endoh, 2026b]. Information Field Geometry Theory (IFGT) describes the complementary informational structure that persists as quasi-closure in the region where  $C$  does not reach unity [Endoh, 2026a].

The present work is positioned as *Application Layer 1* of this framework: it applies the field-theoretic structure of IFGT to the emergence of conceptual landscapes in cognitive and social systems. It does not introduce new foundational variables; it identifies the solution classes of the unified VED×IFGT equation that correspond to cognitive phenomena.

The series architecture is:

| Layer            | Theory                                    |
|------------------|-------------------------------------------|
| L0 (generation)  | VED                                       |
| L0 (structure)   | IFGT                                      |
| L1 (application) | <b>Conceptual Topography [this paper]</b> |
| L1 (application) | Consciousness Theory [forthcoming]        |
| L1 (application) | Intelligence Theory [forthcoming]         |

## 1.3 Main Claims

The paper establishes the following:

1. A **concept** is a locally stable region of the information potential  $\Phi(x, \tau)$ , defined by  $\nabla\Phi(x) \approx 0$  with stability under small perturbations.
2. **Understanding** is a temporary reduction in traversal cost across a conceptual region, not the acquisition of correct meaning.
3. **Thought** is the trajectory of information flow  $J_I$  across the potential landscape; it is constrained by  $\nabla\Phi$  and does not require a directing subject.
4. **Language tokens** function as discrete re-addressable units that couple locally to  $\nabla\Phi$ , deepening attractors through repeated use without carrying semantic content.

5. The strict positivity  $I(x, \tau) > 0$  excludes complete conceptual closure as an ordinary dynamical state and leaves the landscape open to further differentiation.
6. At the **social scale**, the same dynamics produce a shared potential geometry maintained across agents, constituting the structural basis of norms, institutions, and collective intelligence.

## 1.4 Structure of the Paper

Section 2 presents an intuitive picture of conceptual landscapes. Section 3 establishes the formal mapping from VED×IFGT variables to conceptual phenomena. Section 4 derives the core equations governing landscape formation and token–field coupling. Section 5 provides structural interpretation of the main solution classes. Section 6 analyzes the dynamics of sedimentation, forgetting, and conceptual change. Section 7 treats the scale inversion from individual to social landscapes. Section 8 identifies failure modes and limits. Section 9 connects the formal structure to the intuitive descriptions developed in the companion note series. Section 10 discusses implications and open questions. Section 11 concludes.

## 2 Intuitive Picture

Before introducing the formal apparatus, we sketch the intuitive picture that motivates the technical development.

### 2.1 Thought as Movement Across Terrain

Consider the experience of working through a difficult problem. Some directions of thought feel natural and low-cost; others require effort and tend to be abandoned. Some ideas return repeatedly without deliberate intention; others, once visited, leave no trace. The experience of “understanding” something is not the acquisition of a new object, but a change in the ease with which one can move through a region of thought.

These observations suggest a spatial metaphor: thought moves across a terrain whose shape is not fixed in advance but is formed by the history of prior movement. Frequently traversed paths become smoother; regions that are never visited grow inaccessible. The terrain itself is invisible to the traveler, who experiences only the ease or difficulty of each step.

This is the intuitive core of Conceptual Topography. Figure 1 summarizes this geometric picture.

The “terrain” is the information potential  $\Phi(x, \tau)$ , shaped by the accumulated history of information flow  $J_I$ . The “movement” is the trajectory of  $J_I$  along  $-\nabla\Phi$ . The “understanding” is a local reduction in  $|\nabla\Phi|$  in a previously difficult region.

### 2.2 Concepts as Valleys

In this picture, a concept is a valley in the potential landscape: a region of low  $\Phi$  toward which trajectories are naturally drawn. The depth of the valley reflects how frequently the corresponding region has been traversed; its width reflects the range of contexts in which the concept applies.

Crucially, the valley does not contain a meaning. It is a *dynamical attractor*: a region of the field that systematically draws in subsequent trajectories and makes certain patterns of thought more likely. What we call the “meaning” of a concept is, in this framework, the set of trajectories that the attractor makes available.

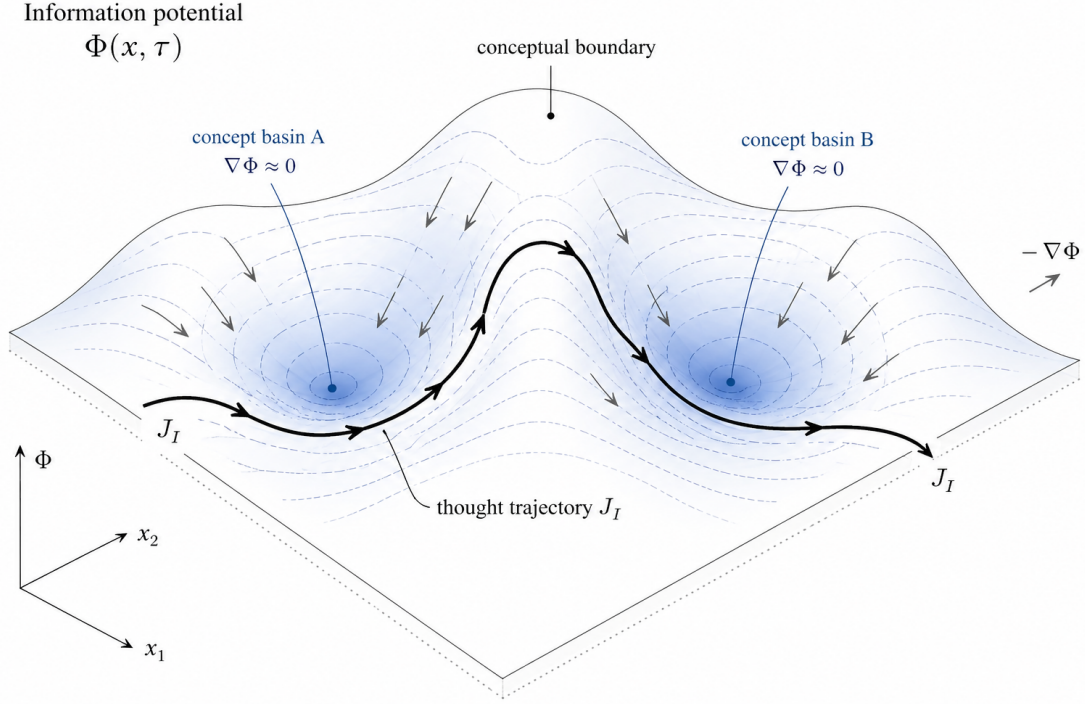


Figure 1: Conceptual landscape as information-potential geometry. The surface represents the information potential  $\Phi(x, \tau)$ . Concepts correspond to locally stable basins where  $\nabla\Phi \approx 0$ . Thought is represented as the trajectory of information flow  $J_I$  constrained by the gradient structure of the landscape. Ridges correspond to high-cost conceptual boundaries.

### 2.3 Linguistic Tokens as Terrain-Shaping Operations

Linguistic tokens — understood broadly to include not only natural language words but any discrete, re-addressable medium capable of acting on the field as a localized perturbation: symbols, emblems, gestures, diagrams, mathematical notation, and so on — do not carry meanings from speaker to listener. They act as localized perturbations that shift the potential landscape slightly toward or away from existing attractors. When a discrete unit is used repeatedly in similar contexts, it deepens the associated valley; when it falls out of use, the valley shallows. Communication succeeds not because meanings are transferred, but because sender and receiver have sufficiently similar landscape geometries that the same perturbation produces similar trajectory shifts.

### 2.4 The Invisible GPU

The landscape operates below the threshold of conscious access. Thought moves across the terrain, but the terrain itself is not represented or inspected during movement. This gives rise to the characteristic phenomenology of cognition: the sense that certain thoughts “come naturally,” that certain connections “suggest themselves,” that understanding arrives suddenly rather than being constructed step by step.

These experiences reflect the geometry of  $\Phi$ , not the content of any represented meaning. The formal development in subsequent sections makes these intuitions precise.

### 3 Formal Mapping

This section establishes the correspondence between VED×IFGT variables and conceptual phenomena. All variables follow the Master Variable Table v2.1.

#### 3.1 The Background Field

The information density field  $I(x, \tau) = f(1 - C(x, \tau))$  provides the continuous substrate on which conceptual structure forms. In this paper,  $I$  is not a new cognitive primitive but a projected realization of the IFGT information-density variable in a cognitive observation window. In the cognitive domain,  $I$  represents the density of unresolved informational tension at position  $x$  and time  $\tau$ : regions where difference chains persist as quasi-closures, neither fully stabilized nor fully dissipated.

High  $I$  corresponds to conceptually active regions; low  $I$  corresponds to regions where structure has largely closed. The strict positivity  $I(x, \tau) > 0$  excludes complete closure as an ordinary state of the conceptual landscape. It does not by itself guarantee a nonzero local gradient at every point; gradients arise from residual asymmetry, nonuniform history, and boundary structure within the generated field.

The information potential

$$\Phi(x, \tau) = (K * I)(x, \tau) \tag{1}$$

is the primary geometric object. It encodes the accumulated bias produced by prior information flow, expressed as a scalar field whose gradient determines the direction of least resistance for subsequent trajectories.

### 3.2 Variable Correspondence Table

| VED×IFGT<br>able                          | vari- | Conceptual domain interpretation                                                                                                                                              | Formal role                                    |
|-------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| $C(x, \tau)$                              |       | Degree of conceptual fixation at $x$                                                                                                                                          | Background field;<br>$C \in [0, C_{\max})$     |
| $I(x, \tau) = f(1 - C)$                   |       | Density of unresolved conceptual tension                                                                                                                                      | Central field; $I > 0$<br>always               |
| $\Phi = K * I$                            |       | The conceptual landscape itself                                                                                                                                               | Potential geometry                             |
| $\nabla \Phi$                             |       | Direction of least cognitive resistance                                                                                                                                       | Constraint force on<br>$J_I$                   |
| $J_I = -D\nabla I + \mu F_I$              |       | Thought as flow                                                                                                                                                               | Trajectory field                               |
| $\sigma = \sigma_d - \sigma_p - \sigma_b$ |       | Rate of landscape deformation                                                                                                                                                 | Source term                                    |
| $\Delta$                                  |       | Novel experience, perception, or linguistic input                                                                                                                             | Driving source                                 |
| $\delta I$                                |       | Unresolved conceptual residue                                                                                                                                                 | Trace that constitutes<br>$I$                  |
| $\tau$                                    |       | Experiential time of conceptual development                                                                                                                                   | Irreversible ordering                          |
| $\ell_i = \mathcal{Q}[I, \Phi, C; \cdot]$ |       | Reusable linguistic token (natural language word, symbol, emblem, gesture, diagram, formula, or any discrete unit capable of acting on the field as a localized perturbation) | Discrete operator on<br>$\Phi$                 |
| $\Phi^*$                                  |       | Mature, sedimented conceptual landscape                                                                                                                                       | Long-time attractor<br>of $\Phi$               |
| $C_{\max}$                                |       | Ideal complete-closure boundary                                                                                                                                               | Boundary representation, not an ordinary state |

**Remark 3.1** (Notational remark). *The operator  $\mathcal{Q}$  is called a quantization operator because its action — extracting discrete, re-addressable units from a continuous field configuration — is structurally isomorphic to quantization in the physical sense: mapping continuous variables to a discrete set of stable values. The term “quantum” is not used as a theoretical claim within CT, but the analogy is preserved in the name of this operator.*

### 3.3 What the Mapping Does Not Claim

The mapping established above does not introduce semantic content, intentionality, or psychological states into the formalism. In particular:

- $\Phi$  is not a meaning field. It is a dynamical constraint geometry that determines which transitions are cheap and which are costly.
- $J_I$  is not thought content. It is a flow direction; the content that appears in experience is a downstream phenomenon.
- $\ell_i$  is not a meaning unit. It is a discrete re-addressable unit that couples locally to  $\nabla \Phi$ .



These exclusions are methodological necessities, not claims about the ultimate nature of meaning or experience.

## 4 Core Equations

### 4.1 Field Evolution

The time evolution of the information density field is governed by:

$$\frac{\partial I}{\partial \tau} + \nabla \cdot J_I = \sigma, \quad (2)$$

where the information flow is

$$J_I = -D\nabla I + \mu F_I, \quad F_I = -\nabla \Phi = \alpha \nabla C, \quad (3)$$

and the generation term decomposes as

$$\sigma = \sigma_{\text{drive}} - \sigma_{\text{dissip}} - \sigma_{\text{barrier}}. \quad (4)$$

The three contributions are:

$$\sigma_{\text{drive}} = a\Delta \left( 1 - \frac{C}{C_*} \right), \quad (5)$$

$$\sigma_{\text{dissip}} = \Gamma C, \quad (6)$$

$$\sigma_{\text{barrier}} = \frac{\kappa_B}{C_{\text{max}} - C}. \quad (7)$$

Here  $C_*$  denotes the effective saturation scale of conceptual quasi-closure within the observation window, while  $C_{\text{max}}$  denotes the ideal complete-closure boundary. In general  $C_* < C_{\text{max}}$  for conceptual regimes.

In the cognitive domain:  $\sigma_{\text{drive}}$  represents the influx of novel input (perception, linguistic perturbation, reward signal);  $\sigma_{\text{dissip}}$  represents the natural decay of unresolved tension (forgetting, habituation);  $\sigma_{\text{barrier}}$  represents a projected or coarse-grained effective resistance against conceptual overclosure. It should not be read as a fundamental external prohibition; it is the cognitive-domain analogue of the effective barrier notation used in VED and IFGT.

### 4.2 Token–Field Coupling

A purely continuous information field cannot sustain stable, low-cost recursion: re-entering a prior field configuration is generically expensive and drift-prone. Stable recursion requires discrete, re-addressable structures that allow the system to re-invoke complex field states at low cost. Figure 2 illustrates this local coupling mechanism.

The medium that enables this structure is the linguistic field, where “linguistic” is understood broadly to include not only natural language but any discrete, re-addressable medium capable of acting on the field as a localized perturbation: symbols, emblems, gestures, diagrams, mathematical notation, and so on.

Linguistic tokens  $\ell_i$  (discrete units drawn from any such medium) couple to the field through the gradient of the potential:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{field}}[I] + \mathcal{L}_{\text{token}}[\ell] + \lambda \sum_i \ell_i \cdot \nabla \Phi. \quad (8)$$

This coupling is purely dynamical: tokens do not encode meanings but bias the local evolution of  $\Phi$  through repeated interaction. Each token reuse reinforces the local gradient, deepening the associated potential well.

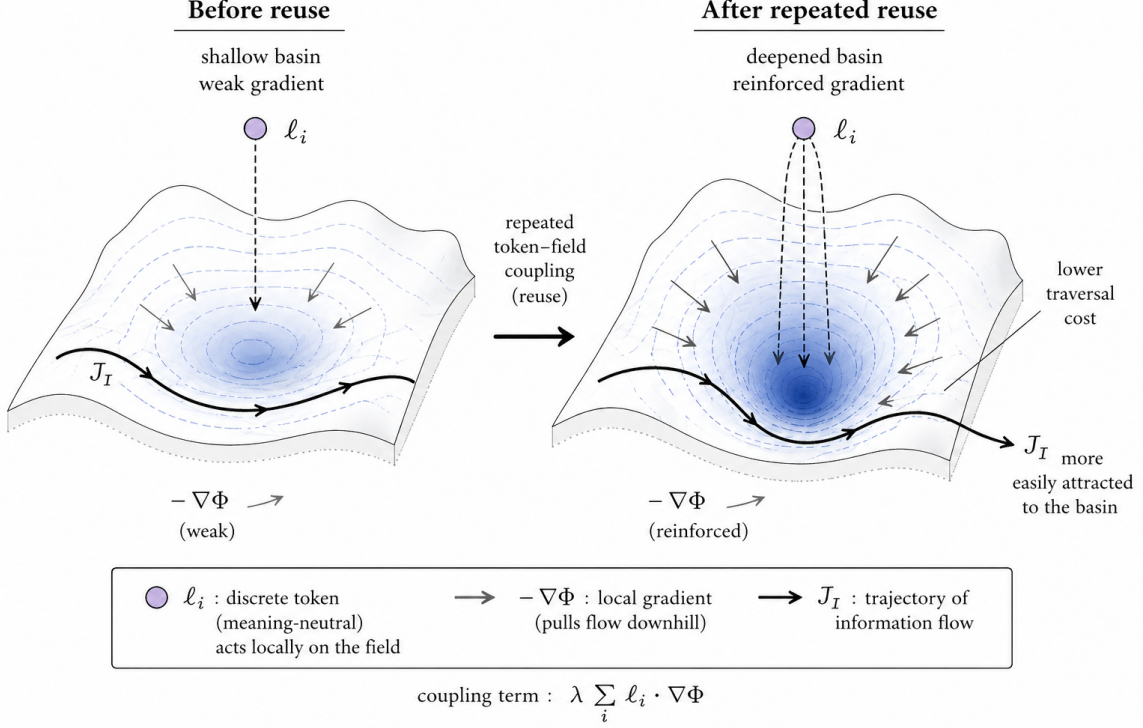


Figure 2: Token–field coupling and basin deepening. Discrete linguistic tokens  $\ell_i$  do not carry semantic content; they act as re-addressable units that couple locally to  $\nabla \Phi$ . Repeated reuse reinforces local gradients, deepens the associated potential basin, and lowers traversal cost for future trajectories of  $J_I$ . The coupling is summarized by the effective term  $\lambda \sum_i \ell_i \cdot \nabla \Phi$ .

### 4.3 Sedimentation and Conceptual Landscape Formation

Under repeated token–field interaction, the potential field approaches a long-time attractor: Figure 3 summarizes the irreversible loop assumed throughout this paper.

$$\Phi(x, \tau) \xrightarrow{\tau \rightarrow \infty} \Phi^*(x). \quad (9)$$

$\Phi^*$  is the sedimented conceptual landscape. It is not necessarily unique or static; it reflects the history of all prior interactions and continues to be reshaped by new input, though at decreasing rate as the landscape matures.

### 4.4 Concept Definition

**Definition 4.1** (Concept). A concept at position  $x$  is a locally stable region of the information potential satisfying:

$$\nabla \Phi(x) \approx 0, \quad \text{with stability under small perturbations of } J_I. \quad (10)$$

The depth of the potential well at  $x$  measures the concept’s resistance to deformation; its width measures its contextual range.

**Definition 4.2** (Understanding). Understanding a conceptual region  $\mathcal{R}$  is the state in which the traversal cost of  $J_I$  through  $\mathcal{R}$  has been reduced to below a system-dependent threshold  $\theta$ :

$$\int_{\mathcal{R}} |\nabla \Phi| dx < \theta. \quad (11)$$

Here  $\theta$  is a system-dependent effective threshold expressing the allowable range of recursive cost. Its concrete value depends on the structure of the system, including  $D$ ,  $\mu$ ,  $\alpha$ , and related parameters, and is

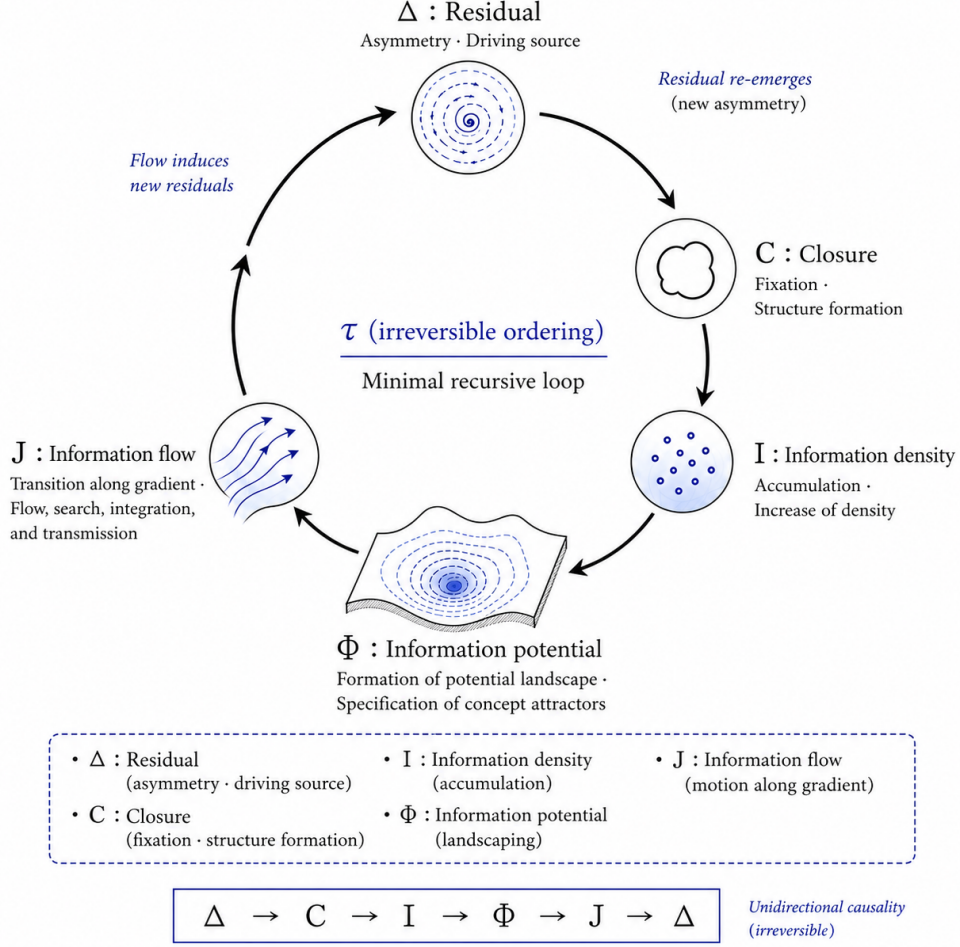


Figure 3: Minimal recursive loop of the VED×IFGT framework. The system evolves through the irreversible cycle  $\Delta \rightarrow C \rightarrow I \rightarrow \Phi \rightarrow J \rightarrow \Delta$ , where residual asymmetry  $\Delta$  drives closure  $C$ , leading to information density  $I$ , formation of potential structure  $\Phi$ , and subsequent information flow  $J$ , which regenerates residuals. The loop is unidirectional and governed by irreversible ordering  $\tau$ .

not specified in the present work. Understanding is not a binary state but a dynamical condition that can be entered and exited as  $\Phi$  evolves.

#### 4.5 The Strict Positivity Constraint

The structural positivity of the information field,

$$I(x, \tau) > 0 \quad \forall x, \tau, \quad (12)$$

follows from  $C < C_{\max}$  combined with  $I = f(1 - C) > 0$  for  $C < C_{\max}$ . Here  $C_{\max}$  denotes the ideal complete-closure boundary, not an ordinary attainable state of the conceptual system.

**Proposition 4.1** (No ordinary complete conceptual closure). *Under the VED×IFGT framework,  $I > 0$  excludes complete conceptual closure as an ordinary dynamical state. In nontrivial generated landscapes with residual asymmetry, spatial inhomogeneity, or boundary structure, this residual positivity generically appears as nonzero gradients in  $\Phi$  and hence as available directions for  $J$ . The proposition should therefore be read as a statement about the non-attainability of final conceptual stasis, not as the pointwise mathematical claim that every positive  $I$  must have  $\nabla \Phi \neq 0$ .*

**Remark 4.1.** *This result has direct implications for theories of intelligence and understanding: no conceptual landscape is finished in the sense of reaching complete closure. Stable attractors may appear locally quiescent, but they remain embedded in a generated field where further differentiation can reappear under new perturbations or wider observation windows.*

## 5 Structural Interpretation

### 5.1 Solution Classes of the Unified Equation

The unified VED $\times$ IFGT equation admits several classes of solutions that correspond to distinct cognitive phenomena. We identify the principal classes here; detailed analysis is deferred to future work.

| Solution class          | Field structure                        | Cognitive interpretation             |
|-------------------------|----------------------------------------|--------------------------------------|
| Smooth gradient region  | $\nabla\Phi$ slowly varying            | Background cognitive disposition     |
| Shallow metastable well | $\nabla\Phi \approx 0$ , small depth   | Tentative or unstable concept        |
| Deep stable well        | $\nabla\Phi \approx 0$ , large depth   | Entrenched concept or belief         |
| Limit-cycle trajectory  | $J_f$ circulates without sedimentation | Recurring thought pattern            |
| Barrier region          | $ \nabla\Phi $ large                   | Cognitive block or resistance        |
| Ridge structure         | Local maximum of $\Phi$                | Conceptual boundary or category edge |

### 5.2 Depth, Width, and Robustness

Three geometric parameters characterize a conceptual attractor:

**Depth** ( $-\min\Phi$  within the well) measures resistance to deformation by new input. Deep concepts are stable under perturbation; shallow concepts are easily displaced.

**Width** (spatial extent of the sub-threshold region) measures contextual range. Broad concepts apply across many domains; narrow concepts are context-specific.

**Robustness** (curvature of  $\Phi$  at the minimum) measures precision. High curvature produces sharp, well-defined concepts; low curvature produces fuzzy or polysemous ones.

These parameters are not fixed intrinsic properties; they evolve continuously with  $\Phi$  under the dynamics of Eqs. (2)–(7).

### 5.3 Conceptual Boundaries

Ridge structures in  $\Phi$  — regions of locally high potential separating neighboring wells — correspond to conceptual boundaries: the distinctions between concepts. These boundaries are not sharp lines but extended regions of high traversal cost.

A key consequence is that conceptual boundaries shift under repeated linguistic or experiential perturbation. Tokens that consistently couple to the gradient near a ridge will either reinforce it (deepening the distinction) or erode it (merging the adjacent concepts). This provides a mechanistic account of semantic drift and conceptual change.

## 5.4 The Vocabulary as a Distribution

The full set of stabilized tokens  $\{\ell_i\}$  constitutes a vocabulary. In this framework, vocabulary is not a list but a distribution over stability measures:

$$\mathcal{V} = \{\ell_i \mid S(\ell_i) > \epsilon\}, \quad (13)$$

where  $S(\ell_i)$  is a stability functional derived from reuse frequency, persistence time, and basin depth, and  $\epsilon$  is a system-dependent threshold.

Vocabulary is a field property, continuously reshaped by use. Token frequencies reflect basin accessibility, not semantic importance. Boundaries between tokens are fuzzy and context-sensitive.

## 6 Dynamics

### 6.1 Sedimentation

Sedimentation is the process by which repeated token–field interaction deepens potential wells and stabilizes conceptual attractors. It proceeds in three stages:

**Stage 1: Initial perturbation.** A novel input  $\Delta$  generates a  $\sigma_{\text{drive}}$  spike, locally increasing  $I$  and creating a transient potential gradient. The information flow  $J_I$  responds by redirecting toward the perturbed region.

**Stage 2: Reinforcement.** If the perturbation is repeated — through subsequent exposure to similar input or through internal recursive activation — the local gradient in  $\Phi$  is reinforced. Token–field coupling (8) ensures that each reuse deepens the associated well.

**Stage 3: Stabilization.** As the well deepens, the traversal cost of the corresponding conceptual region decreases. New trajectories are drawn toward the attractor, further reinforcing it. The concept becomes self-sustaining.

### 6.2 Forgetting

Forgetting corresponds to the relaxation of  $\Phi$  toward a flatter configuration when reuse frequency drops:

$$\sigma_{\text{dissip}} = \Gamma C. \quad (14)$$

This is not erasure of stored symbols but a smooth redistribution of stability across the field. Forgotten concepts do not disappear; their wells shallow, raising the traversal cost until they are effectively inaccessible. Reactivation (“remembering”) corresponds to re-deepening a shallow well through renewed token–field interaction.

### 6.3 Conceptual Change

Conceptual change — the shift in meaning, scope, or application of an established concept — corresponds to a topological change in  $\Phi^*$ : the merger of two wells, the splitting of one well into two, or the displacement of a well from one region of the landscape to another.

These changes require sustained perturbation sufficient to overcome the depth of existing attractors. The effective barrier term  $\sigma_{\text{barrier}}$  plays a protective role here: as  $C \rightarrow C_{\text{max}}$  in a deeply entrenched concept, the projected resistance grows, making radical conceptual change increasingly costly. This should be understood as an effective description of overclosure within the conceptual observation window, not as an externally imposed wall. This accounts for the resistance of deeply held beliefs and intuitions to revision.

## 6.4 Insight as Rapid Topology Change

The phenomenology of insight — the sudden sense that a problem has been resolved or that disparate ideas have been connected — corresponds to a rapid change in the local topology of  $\Phi$ . In field-theoretic terms, this is a bifurcation event: two previously separated wells merge, or a barrier that had prevented flow between two regions suddenly collapses.

The subjective sense of suddenness reflects the threshold character of topological transitions: small perturbations accumulate below the threshold of perceptual change, then produce a discrete topological event when the threshold is crossed.

## 7 Scale Inversion: Individual to Social

### 7.1 The Same Dynamics, Different Dominant Costs

The dynamics described in preceding sections operate at the level of an individual cognitive system. At this scale, the dominant costs are recursion and compression: the system must re-enter prior states at low cost, and must compress high-dimensional continuous field states into reusable discrete handles. Under these constraints, language tokens function as internal control units.

When the same dynamics are extended to a population of interacting agents, the dominant costs shift. At the social scale, synchronization and coordination costs dominate: tokens must stabilize shared trajectories across individuals by reducing uncertainty about each other's landscape geometry.

The shift in dominant costs produces an apparent inversion of linguistic function:

| Scale      | Dominant cost                 | Token function           |
|------------|-------------------------------|--------------------------|
| Individual | Recursion, compression        | Internal control units   |
| Social     | Synchronization, coordination | Shared landscape anchors |

No property of the tokens has changed; only the effective constraint structure changes.

### 7.2 Social Conceptual Landscape

As tokens circulate across a population, each agent's landscape is perturbed by the same linguistic inputs. When agents share sufficiently similar prior landscapes, repeated shared perturbation produces correlated deepening of the same wells: a shared conceptual landscape emerges.

The shared landscape is not stored in any individual agent; it is distributed across the population and maintained through ongoing linguistic interaction. Its geometry is the product of all prior collective token-field interactions.

### 7.3 Norms, Institutions, and Paradigms

Deeply sedimented regions of the social conceptual landscape correspond to norms, institutions, and scientific paradigms. These are not collections of meanings or propositions; they are stable attractor regions of the collective  $\Phi$  whose depth makes deviation dynamically costly.

Institutional stability does not require enforcement by a central authority; it follows from the depth of the collective potential well. Institutional change requires coordinated perturbation sufficient to overcome collective attractor depth — a condition that accounts for the historical difficulty of paradigm shifts.



## 7.4 Intelligence as Landscape Maintenance

At the social scale, intelligence is not a property of individual agents but a distributed process:

*Intelligence is the socially distributed maintenance, synchronization, and gradual reshaping of conceptual landscapes through language.*

Individual agents function as transient operators within this process: they traverse the landscape, introduce local deformations, and transmit stabilized structures to subsequent agents through externalized linguistic logs.

The strict positivity  $I > 0$  means that the social landscape, like the individual landscape, does not reach complete closure as an ordinary dynamical state. Further differentiation and innovation remain possible whenever new perturbations or wider observation windows reactivate residual asymmetries.

## 8 Pathologies and Limits

### 8.1 Deliberate Exclusions

The present framework deliberately excludes the following:

**Semantic theory.** No claims are made about reference, truth conditions, or interpretative content. Meaning, where it appears, is an emergent label applied to stable trajectories; it is not a primitive of the theory.

**Phenomenology and consciousness.** The framework does not generate or explain subjective experience. Conscious access is treated as a downstream phenomenon; its mechanistic basis is the subject of a companion paper on Consciousness Theory.

**Neurobiological implementation.** The theory describes structural and dynamical conditions at a level of abstraction that is independent of specific neural architectures. Correspondence to neural implementation is deferred to future work.

### 8.2 Failure Modes

**Overclosure (too-deep wells).** When the effective  $\sigma_{\text{barrier}}$  grows large relative to  $\sigma_{\text{drive}}$ , existing wells deepen to the point where new input can no longer produce topological change. The landscape hardens; thought becomes repetitive and loses adaptability. This corresponds phenomenologically to rigidity, dogmatism, or entrenched belief.

**Underclosure (too-shallow wells).** When  $\sigma_{\text{dissip}}$  dominates, existing wells shallow faster than new ones can form. The landscape remains flat; thought becomes unstable and fails to accumulate. This corresponds to the failure of concepts to stabilize, preventing the formation of durable understanding.

**Synchronization breakdown (social scale).** When linguistic tokens fail to produce correlated perturbations across agents — due to divergent prior landscapes, insufficient shared exposure, or deliberate manipulation — the social landscape fragments. The result is the loss of shared conceptual terrain, which the framework identifies as the structural basis of communication failure, cultural conflict, and epistemic polarization.

### 8.3 Scope Limits

The theory establishes structural and dynamical conditions; it does not establish:

- Quantitative predictions about specific cognitive systems (this requires concrete specification of  $K, D, \mu, \alpha, \Gamma, \kappa_B$ ).

- Correspondence conditions between  $\Phi$ -geometry and observable neural or behavioral data.
- A complete theory of language acquisition, semantic change, or collective intelligence.

These remain as future work, on the scaffold established here.

## 9 Relation to the Note Series

The companion note series (L1 layer) developed the intuitive picture of conceptual landscapes in accessible prose, using the vocabulary of sedimentation, flow, and terrain. This section makes the correspondence between the note-layer descriptions and the formal structure of the present paper explicit.

| Note-layer description (J: Japanese original)            | Formal correspondence                              |
|----------------------------------------------------------|----------------------------------------------------|
| <i>log no chinden</i> ^ “log sedimentation”              | Deepening of $\Phi$ via $\sigma_{\text{drive}}$    |
| <i>anteiten</i> “stable point”                           | Locally stable region: $\nabla\Phi \approx 0$      |
| <i>gainen no kubomi</i> “conceptual hollow”              | Potential well in $\Phi$                           |
| <i>zentai koubai</i> “overall gradient”                  | Information potential $\Phi = K * I$               |
| <i>tooriyasusa</i> “passability”                         | $-\Phi(x, \tau)$ (high passability = low $\Phi$ )  |
| <i>tani</i> “valley”                                     | Deep well in $\Phi$                                |
| <i>kabe</i> “wall”                                       | Ridge region of high $\Phi$                        |
| <i>rikai shita kankaku</i> “sense of understanding”      | Reduction of traversal cost $\int  \nabla\Phi  dx$ |
| <i>shikou no kuse</i> “habitual thought pattern”         | Limit-cycle trajectory of $J_I$                    |
| <i>GPU-teki na souchi</i> “GPU-like device”              | Sub-threshold operation of $\Phi$ on $J_I$         |
| <i>genko ryuushi</i> “language particle”                 | Discrete token $\ell_i$ coupling to $\nabla\Phi$   |
| <i>asshuku</i> “compression”                             | Quantization operator $\mathcal{Q}$                |
| <i>shakai-teki kyouyu chikei</i> “shared social terrain” | Social $\Phi^*$ maintained across agents           |

### 9.1 What the Note Layer Does Not Claim

The note series deliberately avoided technical formalism and did not commit to specific functional forms. Several descriptions in the note layer are, from the standpoint of the present paper, *interpretations* of field-theoretic quantities rather than definitions.

In particular:

- The note descriptions of “meaning” as an emergent property of repeated traversal correspond formally to the depth of potential wells, not to any semantic object.
- The note descriptions of “understanding” as a change in ease correspond to the traversal cost condition (11).
- The CPU/GPU distinction introduced in the Consciousness note corresponds to the fast-lane / slow-lane decomposition that will be formalized in the companion Consciousness Theory paper.



## 9.2 The L3 → L1 Direction

The present paper (L3) does not revise the note series (L1). The relation is one of support, not replacement: the formal structure established here provides the theoretical basis that stabilizes the intuitive descriptions without overwriting them.

Where the note layer uses metaphor (terrain, flow, sedimentation), the present paper provides the field-theoretic referents. Where the note layer uses phenomenological description (understanding, meaning, thought), the present paper provides structural conditions. The layers are complementary; neither is reducible to the other.

## 10 Discussion

### 10.1 Relation to Existing Frameworks

**Dynamical systems approaches to cognition.** The treatment of concepts as attractors in a dynamical system has precedent in connectionist and embodied cognition research [e.g., [Thelen and Smith, 1994](#)]. The present framework differs in grounding the attractor geometry in a field-theoretic substrate built from the single foundational axiom of VED and the effective dynamical conventions of IFGT, and in providing an explicit account of how discrete linguistic tokens interact with the continuous field.

**Distributional semantics.** Word embedding models place tokens in a geometric space where proximity reflects co-occurrence patterns. The conceptual landscape framework is structurally related but differs fundamentally: in distributional semantics, the geometry is derived from token co-occurrence statistics; here, it emerges from the dynamics of token–field coupling. The geometry reflects the history of *use*, not the statistics of *co-occurrence*.

**Predictive processing and free energy.** Predictive processing accounts treat cognition as the minimization of prediction error [[Friston, 2010](#)]. The present framework is complementary:  $\sigma_{\text{drive}}$  corresponds to prediction error influx, and the deepening of  $\Phi$  wells can be understood as a long-term form of error minimization. However, the present framework does not commit to a Bayesian generative model; the field dynamics are more general.

### 10.2 Open Questions

**Quantitative correspondence.** The framework establishes structural conditions but does not specify the concrete form of the convolution kernel  $K$  or the parameter values  $D, \mu, \alpha, \Gamma, \kappa_B$ . Identifying these from empirical data — behavioral, linguistic, or neural — is a primary open question.

**Topology and concept individuation.** The framework defines concepts as locally stable regions of  $\Phi$ , but the conditions under which a region counts as a single concept rather than a cluster of related concepts are not yet specified. A topological criterion for concept individuation is needed.

**Multi-agent synchronization dynamics.** The social landscape is described qualitatively; a formal treatment of multi-agent token–field interaction, including conditions for stable shared landscape formation and breakdown, remains to be developed.

**Connection to neural implementation.** The correspondence between the field quantities  $I, \Phi, J_I$  and measurable neural quantities (firing rates, connectivity weights, oscillatory dynamics) is a major open question that requires collaboration with empirical neuroscience.

## 11 Conclusion

This paper has developed Conceptual Topography (CT), a field-theoretic account of conceptual landscapes, as the first application layer of the VED×IFGT framework.

The central result is that conceptual structure — the stable geometric configuration that constrains thought without being directly accessible to experience — emerges naturally from the dynamics of the information field  $I(x, \tau)$  under recursive token–field interaction. No semantic primitives, intentional states, or meaning-bearing units are required.

The main structural results are:

1. A concept is a locally stable region of the information potential  $\Phi = K * I$ , defined by  $\nabla\Phi \approx 0$  with stability under perturbation. Its depth, width, and robustness are dynamical quantities that evolve continuously with the history of interaction.
2. Understanding is a reduction in the traversal cost of a conceptual region, not an epistemic achievement. It can be entered and exited as the landscape evolves.
3. Language tokens function as discrete re-addressable units that couple to  $\nabla\Phi$  without carrying semantic content. Their structural role is to enable low-cost recursion and to deepen conceptual attractors through repeated interaction.
4. The strict positivity  $I(x, \tau) > 0$  excludes complete conceptual closure as an ordinary dynamical state. Mature landscapes may appear locally quiescent, but they remain open to further differentiation under new perturbations or wider observation windows.
5. At the social scale, the same dynamics produce a shared conceptual landscape maintained across agents. Norms, institutions, and paradigms are stable attractor regions of the collective potential geometry. Intelligence is the distributed process of maintaining and gradually reshaping this shared landscape.

The framework does not revise or replace the companion note series; it provides the theoretical foundation that stabilizes the intuitive descriptions without overwriting them.

Detailed developments — quantitative parameter identification, topological criteria for concept individuation, multi-agent synchronization dynamics, and neural correspondence — remain as future work on the scaffold established here.

In sum, Conceptual Topography provides a minimal, field-theoretic account of conceptual structure in which thought, understanding, and language emerge as geometric and dynamical properties of the same information field that generates physical structure in the VED×IFGT framework. Within this framework, concepts and physical particles may be read as different projected stabilization regimes of the same generative grammar, not as identical entities or reducible objects. They are distinguished by depth, width, persistence, and observation window, not by a claim that conceptual and physical layers collapse into one another.

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